Lecture plan · for the lecturer

A language model, and changing one

University, scale 2

Time
45 min

Materials
A phone with Darsi; Paper for the rubric line; The slides from the kit

Learning goals

- A guess, not a lookup
- Unless a tool searches

Activity: In the lab

1. Teach one new job
2. Test the old skills too
3. Name what stayed frozen

Discussion

- What did not move?

Exit ticket

- Which part did fine-tuning change here?

Try it in your next lecture
Add a line to the rubric: unnamed weights score zero.

Seminar sheet · for students

A language model, and changing one

University, scale 2

Name _____ Group _____ Date _____

1 **Fine-tuning, in the small lab, changes…**

- ◯ Every weight, starting again from random numbers
- ◯ Only the list of words the model knows
- ◯ The top block, on a few examples

2 **A language model looks facts up as it writes.**

◯ True ◯ False

3 **Match the limit.**

Frozen layers	• • • •	The only part the small lab teaches
Top block	• • • •	A cost of changing a model
Old skill crowded out	• • • •	Weights that did not move

4 **Order a small fine-tune.**

- ☐ Freeze everything but the top block
- ☐ Test the new skill and the old ones
- ☐ Start from a model that already exists
- ☐ Teach the top block with a few examples

Answers · for the lecturer

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1 The top block, on a few examples

Only the top block learns, with the final norm after it; the vocabulary, the embedding, the output layer and the lower blocks stay frozen.

2 False

It guesses the next token from patterns, unless a separate tool searches.

3 Frozen layers — Weights that did not move
Top block — The only part the small lab teaches
Old skill crowded out — A cost of changing a model

Say which weights moved before you claim the model learned a job.

4 Top to bottom: 2, 4, 1, 3

Freezing keeps the change small: only the top block moves, so a phone can do it and the lower blocks keep what they knew.

For you to think about

What sentence will score zero?